

I2C Task Code

```
/*  
 * File:   main.c  
 * Project: I2C EEPROM Interfacing  
 * Author: Sagar  
 */  
  
#include <xc.h>  
#include "config.h"  
#include "I2C_EEPROM.h"  
#define _XTAL_FREQ 16000000  
  
void main(void)  
{  
  
    I2C_Master_Init(100000);    // Initialize I2C Master with 100kHz clock  
    TRISD = 0x00;               // Set PORTD as output  
  
    // Write Operations  
    EEPROM_Write(0x23, 'A');    // Write 'A' (0x41) to address 0x0023  
    EEPROM_Write(0x28, 'B');    // Write 'B' (0x42) to address 0x0028 PORTD = 01000010  
    EEPROM_Write(0x36, 'C');    // Write 'C' (0x43) to address 0x0036  
    __delay_ms(10);             // EEPROM write delay  
  
    PORTD = 0x00;
```

```
PORTD = EEPROM_Read(0x23); // Read 'A' (0x41) from 0x0023 → RD6 & RD0 ON  
__delay_ms(3000);
```

```
PORTD = EEPROM_Read(0x28); // Read 'B' (0x42) from 0x0028 → RD6 & RD1 ON  
__delay_ms(3000);
```

```
PORTD = EEPROM_Read(0x36); // Read 'C' (0x43) from 0x0036 → RD6 & RD1 & RD0 ON  
__delay_ms(3000);
```

```
while(1) {  
    // while loop  
}  
return;  
}
```

I2C Debugger Output

